Utkarsh Anshu Gupta

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Education

University of Washington — Seattle, WA
Master of Science, Data Science — *Sep'26 – Jun'28*

Indian Institute of Technology, Bombay — Mumbai, India
B.Tech Chemical Engineering, Minor in Computer Science — *Oct'20 – May'24*

Technical Skills

Machine Learning: XGBoost, PyTorch, Scikit-learn, Transformers, OpenCV, SBERT
Languages: Python, SQL, MATLAB, R, C++, LaTeX
LLM & Agentic AI: LangGraph, OpenAI Agents SDK, RAG, LoRA
Data & MLOps: ETL/ELT, Databricks, PySpark, PostgreSQL, Azure, Azure DevOps, Docker, CI/CD, Git

Professional Experience

Air India, Tata Group | Data Scientist | Digital and Technology — Nov'24 – Sep'26

Enterprise Flight Operations Analytics Chatbot

- Architected & deployed a **GPT-4**-powered conversational analytics chatbot on MS Teams using Azure & PostgreSQL
- Achieved **8** second end-to-end latency by designing an NL2SQL pipeline for natural-language flight operations analytics
- Developed **65+ OWASP**-aligned LLM security guardrails for prompt injection and sensitive data leakage prevention
- Secured LLM interactions for **100+** non-technical users using dual-layer input/output sanitization across all requests

Crew Feedback Intelligence Pipeline

- Built an agile pipeline using **RoBERTa-MNLI** with bi-encoder retrieval, re-ranking, & semantic similarity matching
- Matched frontier LLM performance with **97%** accuracy across **10** behavioral dimensions at **zero** additional API cost
- Engineered an end-to-end MLOps pipeline on Databricks using PySpark, Azure DevOps, & automated ingestion triggers
- Scaled processing to **100k+** entries, reducing manual processing time from days to a few hours through automation

Route-Level Demand Forecasting

- Built a **PySpark** feature engineering pipeline for **1.1M+** records, incorporating spatial, statistical, and holiday signals
- Reduced forecast error to **5% MAPE** through Bayesian hyperparameter optimization for route-level seat estimation
- Designed an autoencoder for **cold-start** forecasting by learning city embeddings from demographic & economic signals

Reliance Industries Limited | Technical Intern | CTS Department — May'23 – Jul'23

- Developed an ASPEN-based process optimization model using simulated & plant operating data for sensitivity analysis
- Delivered a **50%** increase in cryogenic ethylene throughput while maintaining fixed downstream operating conditions

Publications

Hyperspectral Unmixing Algorithms with Explainability: An Assessment of Geometric and Deep Learning Methods
Utkarsh Anshu Gupta, Soumyashree Kar | Oral Presentation | Accepted, IEEE IGARSS 2026

Research Experience

ML Techniques for Hyperspectral Unmixing — Jun'25 – Feb'26

Guide: Prof. Soumyashree Kar | CSRE, IIT Bombay

- Benchmarked **63** hyperspectral unmixing algorithms across **4** HSI datasets using an automated evaluation framework
- Reduced endmember reconstruction error by 28%, closing 87% of the gap to SOTA using DDPM-based augmentation
- Evaluated **34** model checkpoints using **Hungarian-matched** evaluation, ensuring correct endmember comparisons
- Built a config-driven PyTorch framework supporting **3** architectures across **4** datasets for reproducible experiments

Data-Driven Discovery of High-Performing Phosphine Ligands — Aug'24 – Apr'25

Guide: Prof. Debabrata Maiti | Chemistry, IIT Bombay

- Trained **190** XGBoost models achieving median R^2 of **0.86** and maximum R^2 of **0.96** for DFT descriptor prediction
- Predicted quantum chemical descriptors from **1024-bit** Morgan fingerprints, enabling phosphine ligand screening
- Engineered a domain-specific feature selection pipeline via **threshold optimization** across selected DFT descriptors

Startup Experience

NYX | AI Engineer | IMG – IMG Generation — Jul'24 – Aug'24

- Developed an IMG-IMG generation pipeline using **JuggernautV9** to reduce total customer acquisition cost by **30%**
- Preprocessed images using BRIA-RMBG - 1.4 for background segmentation and RealVisXL_V4.0 model for masking

Feynn Labs | DS Engineer | Supply Chain Optimization — Jun'24 – Jul'24

- Achieved an RMSE of **0.94** through XGBoost-based daily demand forecasting and **ABC** product segmentation
- Designed a Google OR-Tools **route** optimization engine with an interactive **Plotly** dashboard for delivery planning